Nuskha — Prescription and Discharge Instruction Agent

Simple Problem Formulation and Environment Classification

1. Problem Formulation

Element	Description
Initial State	The agent receives a photo of a handwritten or printed prescription or hospital discharge slip, along with a short patient profile giving age, known allergies, existing medicines, language preference and usual meal times. Nothing on the slip has been read yet.
Actions	Preprocess and OCR the image; break each line into medicine name, strength, frequency, route and duration; match the brand name to its generic salt; score the confidence of every extracted field; hold back any field below the threshold; check for duplicate salts, interactions and allergy conflicts; convert frequency codes into real clock times; generate plain-language instructions in Urdu or English; answer follow-up questions from the parsed record.
Transition Model	Each stage updates the structured record. The image becomes text, the text becomes fields, the fields become identified medicines with confidence scores, and the confident ones become a timed schedule. Details entered in the patient profile update the safety alerts, and a confirmed field moves from the unclear list into the schedule.
Path Cost	Cost can be measured by processing time per slip, the number of OCR passes needed, and the number of fields that end up unread. The objective is a complete, correct schedule with the fewest unread fields and no wrong readings, since a wrong reading is far more costly than an unread one.
Goal Test	The patient or caregiver has a day-by-day schedule with clock times, a one-line explanation of each medicine, the food and timing rules, the red-flag symptoms and the follow-up date. Every line is traceable to a line on the slip, and anything unclear is listed for confirmation rather than guessed.
State Space	All possible combinations of uploaded images, OCR output, extracted fields, confidence scores, identified salts, safety alerts, generated schedules, unread items and caregiver questions.
Search Space	All possible sequences of preprocessing, reading the image, extracting fields, matching medicines, applying the confidence gate, running safety checks, building the schedule, generating the explanation and answering questions.
Agents	Single agent. One agent reads the slip and explains it to the patient. The doctor and pharmacist remain outside the system as human authorities the agent refers the user back to.

2. Environmental Properties

Property	Classification	Why
Observability	Partially Observable	The agent only sees the slip and whatever the patient declares. It cannot see the doctor's reasoning, the patient's full medical history, or any medicine being taken at home that was not mentioned.
Determinism	Stochastic / Non-deterministic	Handwriting OCR is uncertain. Lighting, photo quality and writing style change the result, so the same slip can be read differently on two attempts.
Episodes	Sequential	What the confidence gate accepts at stage four decides what reaches the schedule, and every later answer is based on the record built earlier in the pipeline.
Change	Dynamic	New slips, updated allergy information, confirmed fields from the pharmacist and fresh caregiver questions keep arriving, and each one changes the stored record and the output.
Values	Discrete & Continuous	Medicine names, frequency codes and number of days are discrete, while OCR confidence scores, dose strengths and dosing times are continuous.
Agents	Single Agent	One agent serves one patient or caregiver at a time, with no competing or cooperating agent inside the system.

3. Choosing the State Type

Option	Suitability	Explanation
Single-state	Not suitable	The agent does not know what is written on the slip before reading it, and the outcome of the OCR step cannot be predicted in advance.
Multi-state	Most suitable	The agent passes through many states: raw photo, OCR text, extracted fields, identified medicines, confidence-filtered set, safety alerts, timed schedule and the final plain-language instructions.

Final Answer: Nuskha is best classified as a **Multi-state** agent, because it moves through several changing states while reading the slip, extracting and identifying the medicines, filtering out what it cannot read confidently, and building the schedule and instructions on top of that. The environment is **partially observable** and **stochastic**, since the agent only sees the slip and the declared profile, and handwriting recognition can never be guaranteed correct — which is why unclear fields are reported for confirmation instead of being guessed.